



End Semester Exam Calculus-I (MA1110)

Full Marks: 60
Time- 2 hours

Solve all the questions:

1. Let $a_{n+1} = a_n^2$, $\forall n$ and $0 < a_1 < 1$. Then $\{a_n\}$ is

[2 marks]

- a) null sequence
b) divergent sequence
c) monotone increasing sequence
d) None of these

2. If $a_n = \begin{cases} 1 + \frac{1}{n}, & n \text{ is even} \\ -1 - \frac{1}{n}, & n \text{ is odd} \end{cases}$, then

[2 marks]

- a) $\limsup_{n \rightarrow \infty} a_n = \liminf_{n \rightarrow \infty} a_n = 1$
b) $\limsup_{n \rightarrow \infty} a_n = \liminf_{n \rightarrow \infty} a_n = -1$
c) $\limsup_{n \rightarrow \infty} a_n = 1, \liminf_{n \rightarrow \infty} a_n = -1$
d) None of these

3. Let $\{S_n\}$ is the unbounded above sequence of positive real numbers, where $S_n = \sum_{k=1}^n A_k$, $n = 1, 2, 3, \dots$, $A_k > 0$, and $\{T_n\}$ is the sequence defined as $T_n = \sum_{k=1}^n \frac{A_k}{S_{k-1}S_k}$, $n = 2, 3, \dots$.

Then $\lim_{n \rightarrow \infty} T_n$

- a) $\frac{1}{A_1}$
b) 0
c) $A_1 + A_2$
d) ∞

$\frac{1}{S_1} - \frac{1}{S_2} = \frac{A_2}{S_1 S_2}$
 $\frac{1}{S_2} - \frac{1}{S_3} = \frac{A_3}{S_2 S_3}$
 $\vdots$
 $\frac{1}{S_{n-1}} - \frac{1}{S_n} = \frac{A_n}{S_{n-1} S_n}$
[2 marks]

4. Let $b_n = 2 - \frac{(-1)^n}{n}$ and $a_n = b_n - b_{n+1}$, $\forall n \in \mathbb{N}$, then the series $\sum_{n=1}^{\infty} a_n$

[2 marks]

- a) converges to 2
b) diverges
c) converges to $\frac{1}{2}$
d) None of these

5. If $\sum_{n=1}^{\infty} a_n$ converges, then the limit of the sequence $\left\{ \frac{a_n - 1}{n^n} \right\}_{n \in \mathbb{N}}$ is

[2 marks]

- a) 1
b) 0
c) -1
d) ∞

6. The sum of the series $\frac{1}{3} + \left(\frac{1}{3}\right)^{1+\frac{1}{2}} + \left(\frac{1}{3}\right)^{1+\frac{1}{2}+\frac{1}{4}} + \dots$ equals to

[2 marks]

- a) $\frac{1}{9}$
b) $\frac{1}{3}$
c) 2
d) None of these

$\left(\frac{1}{3}\right) \left(1 + \frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots \right)$

7. Let $f(x) = \begin{cases} x^2, & \text{when } 0 \leq x < \frac{1}{2} \\ \frac{1}{4}, & \text{when } x = \frac{1}{2} \\ 1 - x^2, & \text{when } \frac{1}{2} < x < 1 \end{cases}$

Then f is

- a) continuous at $x = \frac{1}{2}$
b) discontinuous at $x = \frac{1}{2}$

$(-1)^{n+1} \cdot \left[\frac{1}{n} + \frac{1}{n+1} \right]$

$2 - \frac{(-1)^n}{n} - \left(2 - \frac{(-1)^{n+1}}{n+1} \right)$
[2 marks]

c) $\lim_{x \rightarrow \frac{1}{2}} f(x)$ exists

d) $\lim_{x \rightarrow \frac{1}{2}^+} f(x) = f\left(\frac{1}{2}\right)$

8. Let $f(x) = \begin{cases} \frac{1 - \cos x}{x^2}, & \text{for } x \neq 0 \\ 1, & \text{for } x = 0 \end{cases}$

[2 marks]

Then

a) $\lim_{x \rightarrow 0} f(x)$ does not exist

b) f is continuous at $x = 0$

c) f has discontinuity at $x = 0$

d) None of these

9. Let $f(x) = \begin{cases} x^{\frac{1}{3}}, & \text{for } -1 \leq x \leq 0 \\ -x^{\frac{1}{3}}, & \text{for } 0 < x \leq 1 \end{cases}$

[2 marks]

Then

a) $f'(0) = 0$

b) Rolle's Theorem is not applicable in $[-1, 1]$

c) f is not continuous at $x = 0$

d) Rolle's Theorem is applicable in $[-1, 1]$

10. Let $f(x) = |x^3|$, $\forall x \in \mathbb{R}$, then the function $f(x)$ at $x = 0$ is

[2 marks]

a) continuous but not differentiable

b) continuous and once differentiable only

c) twice differentiable but not thrice

d) thrice differentiable

11. Let $f(x)$ and $g(x)$ be differentiable on $0 \leq x \leq 2$ such that $f(0) = 2$, $f(2) = 5$, $g(0) = 0$ and $f'(x) = g'(x) \forall x \in (0, 2)$ and $g'(x) \neq 0$, then the value of $g(2)$ is

a) -1 b) 1 c) 2 d) 3

[2 marks]

12. Let $S_n = \int_0^1 \frac{n x^{n-1}}{1+x} dx$, for $n \geq 1$. Then as $n \rightarrow \infty$, the sequence $\{S_n\}$ tends to

a) 0 b) $\frac{1}{2}$ c) 1 d) ∞

[2 marks]

13. Which one of the following is true?

[2 marks]

a) Every sequence that has a convergent subsequence is a Cauchy sequence

b) Every sequence that has a convergent subsequence is a bounded sequence

c) Then sequence $\{\sin n\}$ has a convergent subsequence

d) The sequence $\{n \cos \frac{1}{n}\}$ has a convergent subsequence

14. If $f(x) = \begin{cases} 1+x, & \text{for } x < 0 \\ (1-x)(ax+b), & \text{for } x \geq 0 \end{cases}$ satisfies the assumptions of Rolle's theorem in the interval $[-1, 1]$, then the ordered pair (a, b) is

Page-2

$\int u \cdot v \, dv$

$\int u \cdot v \, dv$

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$u \cdot v - \int u \cdot v \, dv$

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- a) $(2, -1)$ b) $(-2, -1)$ c) $(-2, 1)$ ~~d) $(2, 1)$~~ [2 marks]

15. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies the relation $f(x+y) = f(x) \cdot f(y) \quad \forall x, y \in \mathbb{R}$ and

$f(x) \neq 0, \quad \forall x \in \mathbb{R}$. If f is differentiable at 0 and $f'(0) = 2$, then [2 marks]

- a) $f'(x) = 0 \quad \forall x \in \mathbb{R}$ b) $f'(x) = f(x) \quad \forall x \in \mathbb{R}$
 c) $f'(x) = 3f(x) \quad \forall x \in \mathbb{R}$ ~~d) $f'(x) = 2f(x) \quad \forall x \in \mathbb{R}$~~

16. Find the conditions for $c > 0$, when the sequence $\{x_n\}$ defined by $x_1 = c, \quad x_{n+1} = \frac{6(1+x_n)}{7+x_n}$

is monotone increasing and monotone decreasing. Check the convergence of the sequence

$\{x_n\}$ and find its limit.

[6 Marks]

17. Investigate the convergence and divergence of the following series

$$\frac{1^2}{2^2} + \frac{1^2 \cdot 3^2}{2^2 \cdot 4^2} x + \frac{1^2 \cdot 3^2 \cdot 5^2}{2^2 \cdot 4^2 \cdot 6^2} x^2 + \frac{1^2 \cdot 3^2 \cdot 5^2 \cdot 7^2}{2^2 \cdot 4^2 \cdot 6^2 \cdot 8^2} x^3 + \dots, \text{ for } x \in (0, \infty) \setminus \{1\}. \quad [6 \text{ Marks}]$$

18. Find the sum of the following series if it is convergent

$$\left(\frac{1}{2 \cdot 3} + \frac{1}{2^2 \cdot 3}\right) + \left(\frac{1}{2^2 \cdot 3^2} + \frac{1}{2^3 \cdot 3^2}\right) + \left(\frac{1}{2^3 \cdot 3^3} + \frac{1}{2^4 \cdot 3^3}\right) + \dots \quad [6 \text{ Marks}]$$

19. Use sequential criterion for the limit to evaluate the following limit

$$\lim_{x \rightarrow \infty} x^2 \operatorname{sgn}(\cos x), \text{ where } \operatorname{sgn}(x) \text{ is the sign or signum function.} \quad [6 \text{ Marks}]$$

20. (i) Find the subset of $\mathbb{R}$, where the function $f(x) = |x+1| + |x-1|, \quad \forall x \in \mathbb{R}$, is continuous and differentiable. [2 Marks]

$$(ii) \text{ Let } f(x) = \begin{cases} x^{\frac{3}{2}} \sin \frac{1}{x}, & \text{if } x \in \mathbb{R} \setminus \{0\} \\ 0, & \text{if } x = 0. \end{cases} \quad [4 \text{ Marks}]$$

Is the function $f(x)$ differentiable at $x = 0$? Justify your answer.

*****Best of Luck*****